

# COHORT RETENTION ANALYSIS

Management Report

|              |                                                               |
|--------------|---------------------------------------------------------------|
| Prepared for | CRM / Marketing Operations, Category Managers, Analytics Team |
| Prepared by  | Mohammad Ammar — Apex AnalyticX                               |
| Engagement   | E-Commerce Customer Intelligence Series — Project 3 of 4      |
| Date         | July 2026                                                     |

## 1. Objective

Provide the operational detail needed to design, prioritize, and measure a retention intervention — translating the executive-level findings into specific targeting logic, timing windows, and tracking KPIs that CRM/Marketing Ops can act on directly.

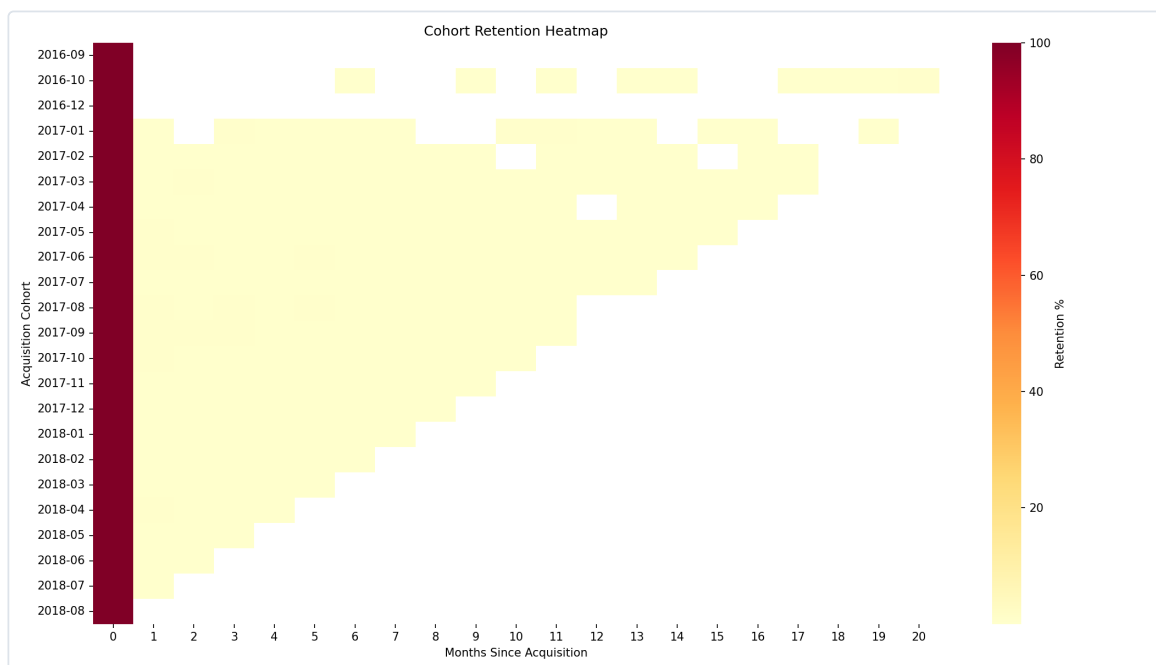
## 2. Dataset & Scope

| Item               | Detail                                                                                                    |
|--------------------|-----------------------------------------------------------------------------------------------------------|
| Source             | Olist Brazilian E-Commerce dataset (reused from Revenue Analysis and RFM Segmentation projects)           |
| Customers analyzed | ~95,560 unique customers ( <code>customer_unique_id</code> )                                              |
| Time window        | September 2016 – August 2018                                                                              |
| Cohorts            | 23 monthly acquisition cohorts                                                                            |
| Order filter       | Valid orders only ( <code>order_status</code> excludes <code>canceled</code> , <code>unavailable</code> ) |
| Cohort definition  | Customer's earliest <code>order_purchase_timestamp</code> → assigned cohort month                         |
| Retention metric   | <code>active_customers(month N) / cohort_size(month 0)</code>                                             |

Full field-level detail is in `docs/data_dictionary.md`; full construction logic is in `docs/09_Data_Model.md`.

### 3. Full Retention Pattern

| Metric                                                   | Value                                                                                         |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Average month-1 retention (mature cohorts, Jan–Jun 2017) | <b>0.33%</b>                                                                                  |
| Best-performing cohort (month-1)                         | August 2017 — <b>0.65%</b>                                                                    |
| Worst-performing cohort (month-1)                        | July 2018 — <b>0.099%</b>                                                                     |
| Retention, months 1–2 (relative peak)                    | 0.33% – 0.36%                                                                                 |
| Retention, months 3–8 (decline phase)                    | 0.20% – 0.32%                                                                                 |
| Retention, month 13+ (floor)                             | ~0.15%, with sporadic single-digit-customer upticks (sample noise, not genuine re-engagement) |



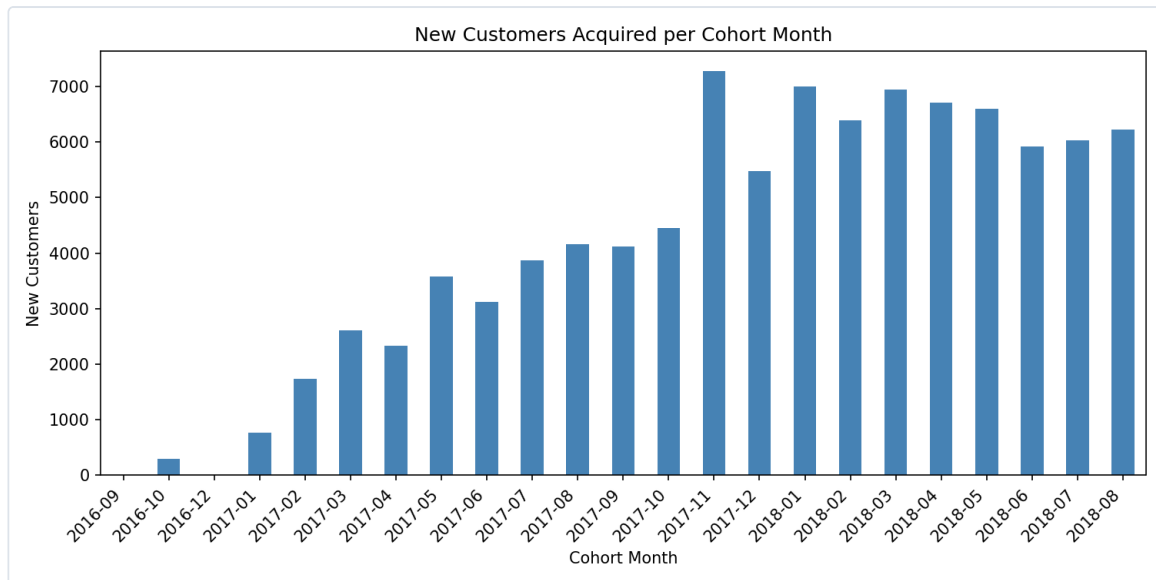
*Full cohort × month retention matrix — near-zero outside the month-0 column across all 23 cohorts.*

**Read:** the strongest — and only meaningfully actionable — retention window is **months 1–2 post-purchase**. By month 3, the opportunity has already started to erode; by month 13, it has flattened at a residual floor that is unlikely to represent recoverable behavior without a fundamentally different intervention (e.g., win-back campaigns rather than standard re-engagement).

### 4. Cohort Volume Context

| Period | Customers Acquired |
|--------|--------------------|
| 2017   | 43,468             |
| 2018   | 51,792             |

| Cohort Size Reference Point | Value                    |
|-----------------------------|--------------------------|
| Smallest cohort (Jan 2017)  | 762 customers            |
| Typical 2018 cohort         | 6,000 – 7,000+ customers |
| Approximate growth multiple | ~8x                      |



*Cohort acquisition volume climbing steadily while retention rate stays flat.*

**Operational implication:** acquisition volume scaling did not move the retention needle — this rules out "wait for scale to fix it" as a viable strategy and confirms a targeted, designed intervention is required. It also means the addressable audience for a re-engagement pilot is now large enough (thousands of first-time buyers per month) to run a statistically meaningful test quickly.

## 5. Recommended Intervention Design

### 5.1 Targeting Window

**Day 15 – Day 45** post-first-purchase. This sits inside the relative-peak retention window (months 1–2) identified in Section 3, while giving enough lead time post-delivery for a genuine second-purchase consideration moment (as opposed to triggering before the first order has even arrived).

### 5.2 Suggested Campaign Structure

| Touchpoint | Timing | Channel                             | Intent                                                                       |
|------------|--------|-------------------------------------|------------------------------------------------------------------------------|
| Touch 1    | Day 15 | Email                               | Soft nudge — "how was your order," light product recommendation, no discount |
| Touch 2    | Day 30 | Email + on-site banner if logged in | Incentive introduced (discount/free shipping threshold)                      |
| Touch 3    | Day 45 | Email (final)                       | Urgency framing on the incentive, last touch before window closes            |

*(Exact incentive structure/economics should be validated against margin data by Finance before launch — out of scope for this analysis.)*

### 5.3 Pilot Design

- **Test group:** all first-time buyers from the next 2–3 cohort months
  - **Control group:** matched cohort(s) receiving no intervention, or a randomized holdout within the same cohort month
  - **Primary success metric:** month-1 retention rate, test vs. control
  - **Minimum viable lift to declare success:** any statistically significant increase above the 0.33% mature-cohort baseline; a lift toward 1.5–2% would represent the 4–6x improvement referenced in the Executive Report
- 

## 6. Root-Cause Research Workstream (Parallel Track)

Recommended inputs to prioritize, in order of likely diagnostic value:

1. **Delivery performance data** — late/on-time delivery rates, damage/return rates, by cohort and category
2. **Seller-level consistency** — repeat-purchase rate cut by seller, to isolate whether specific sellers are dragging down the average
3. **Support ticket themes** — qualitative review of post-purchase complaints/contact reasons
4. **Existing CRM/campaign history** — confirm whether any prior re-engagement outreach has ever been sent, and if so, why it didn't move retention

This workstream directly informs whether Recommendation 2 from the Executive Report should focus on logistics, seller quality, product/category mix, or platform UX.

---

## 7. KPIs to Track Going Forward

| KPI                             | Definition                                                              | Query Reference                             |
|---------------------------------|-------------------------------------------------------------------------|---------------------------------------------|
| Month-1 retention rate          | % of cohort returning within 1 month of first purchase                  | sql/<br>03_kpi_queries.sql , KPI<br>3       |
| Month-3 retention rate          | % of cohort returning within 3 months                                   | Extend KPI 3 pattern to<br>month_number = 3 |
| Repeat purchase rate (all-time) | % of customers with 2+ distinct order months, ever                      | sql/<br>03_kpi_queries.sql , KPI<br>1       |
| Median time to second purchase  | Median days between order 1 and order 2, for repeat customers           | sql/<br>03_kpi_queries.sql , KPI<br>2       |
| Cohort acquisition volume       | New customers per cohort month, for cross-reference with retention rate | sql/<br>03_kpi_queries.sql , KPI<br>4       |

**Recommendation:** these five KPIs should be added to the standing executive/marketing dashboard as a dedicated "Retention" section, refreshed monthly, cohort by cohort — not buried in a one-time report.

## 8. Data Caveats for This Workstream

- `customer_unique_id` may not perfectly capture repeat behavior if the same customer used multiple accounts or emails — treat the 0.33% figure as a reasonable floor estimate, not necessarily an exact one
- No marketing spend or historical campaign data was available in this dataset — the pilot design in Section 5 assumes no prior re-engagement outreach has been tested; confirm this before launch
- Seasonality has not been fully isolated from genuine retention trend in this pass — if the root-cause workstream (Section 6) surfaces strong seasonal patterns, revisit the cohort comparisons with seasonal adjustment

## 9. Supporting Technical Artifacts

| Artifact                      | Location                           |
|-------------------------------|------------------------------------|
| Cohort assignment logic       | sql/01_cohort_assignment.sql       |
| Retention matrix construction | sql/02_cohort_retention_matrix.sql |
| KPI queries                   | sql/03_kpi_queries.sql             |
| Data cleaning pipeline        | python/01_data_cleaning.py         |
| Cohort construction (Python)  | python/02_cohort_construction.py   |
| EDA / visualization           | python/03_eda_visualization.py     |
| Full KPI definitions          | docs/11_KPI_Definitions.md         |
| Data dictionary               | docs/data_dictionary.md            |

*Prepared by Mohammad Ammar — Apex AnalyticX — E-Commerce Customer Intelligence Series.  
This report is intended for operational use by CRM, Marketing, and Analytics teams executing on the Executive Report's recommendations.*